

-
- `docker run -itd --name my-ml-container --gpus all -v $(pwd):/workspace pytorch/pytorch:2.2.0-cuda12.1-cudnn8-runtime bash`
 - `docker ps`
 - `docker start my-ml-container`
 - `docker exec -it my-ml-container bash`
 - `docker stop my-ml-container`
 - `docker rm my-ml-container`
 - `docker run -itd --name my-ml-container --gpus all -v $(pwd):/workspace _p
8501:8501 pytorch/pytorch:2.2.0-cuda12.1-cudnn8-runtime bash`

STREAMLIT

MLOPS/DLOPS LAB

WHAT IS STREAMLIT

- Streamlit is a Python library that lets you build web apps using only Python — no HTML, CSS, or JavaScript needed.
- Python script ---> Interactive Web app.

STEP – I: INSTALL STREAMLIT

```
pip install streamlit
```

STEP – 2 : RUN STREAMLIT

```
streamlit run app.py
```


STEP – 3 : APP FILE

- Create a file app.py and write content there:

```
import streamlit as st

st.title("My First Streamlit App")

st.write("Hello class! 🙋")

name = st.text_input("Enter your name")

if name:
    st.write(f"Welcome, {name}!")
```

OUTPUT

My First Streamlit App

Hello class! 🙋

Enter your name

Muskan

Welcome, Muskan!

TEXT ELEMENTS

```
st.title("Title")
st.header("Header")
st.subheader("Subheader")
st.text("Simple text")
st.markdown("**Bold Text**")
st.write("Can print almost anything")
```

Title

Header

Subheader

Simple text

Bold Text

Can print almost anything

USER INPUTS

```
age = st.slider("Select your age", 0, 100, 25)
number = st.number_input("Enter a number")
choice = st.selectbox("Choose option", ["A", "B", "C"])
check = st.checkbox("Click me")
button = st.button("Submit")
```

OUTPUT

Select your age

14

0 100

Enter a number

0.07 - +

Choose option

C v

☒ Click me

Submit

VISUALIZE DATA

```
import pandas as pd

df = pd.DataFrame({
    "Name": ["A", "B", "C"],
    "Score": [90, 85, 88]
})

st.dataframe(df)
```

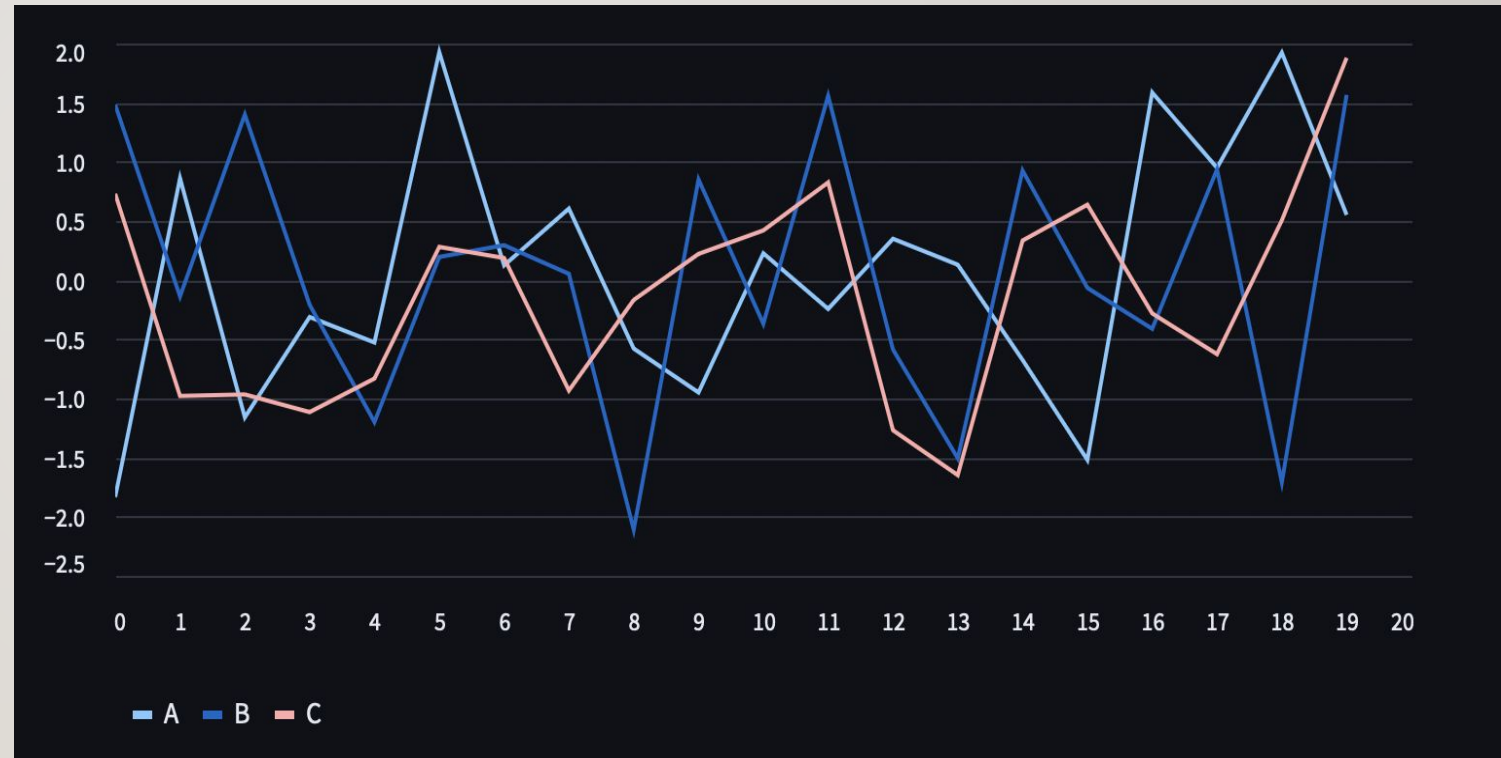
	Name	Score
0	A	90
1	B	85
2	C	88

VISUALIZE DATA

```
import numpy as np

chart_data = pd.DataFrame(
    np.random.randn(20, 3),
    columns=['A', 'B', 'C']
)

st.line_chart(chart_data)
```



STATE SESSION

```
if "counter" not in st.session_state:  
    st.session_state.counter = 0  
  
if st.button("Increase"):  
    st.session_state.counter += 1  
  
st.write("Counter:", st.session_state.counter)
```


RUN DL CODE DEMO

MNIST CLASSIFICATION



OTHER FEATURES

- Visualize equations
- Visualize code blocks
- Input variations – from simple text to dropdown to camera option
- Data visualization – Pandas, numpy, Lists and dicts
- Chart visualization
- Layout variations – sidebar, container
- Media support – image, text, video ..

OTHER ADVANCED FEATURES

- State session
- Caching and performance
- Multipage apps
- Custom themes
- Deployment
- Forms

THANKYOU!

MNIST CLASSIFICATION